CONVERSATION INTELLIGENCE · CONSUMER BANK

# Companion for CallRadar & the Quality Checks

Turn 1,441 raw support-call recordings into a manager's dashboard — who called, what they wanted, how their mood moved, whether it was *really* resolved, which calls need attention today — and **the exact moment on the call that proves every judgment.**

Calls `1,441` | Faithfulness `99.9%` | Diarization `100%` | Resolution-risk `195` | API keys needed `0`

You get audio, not transcripts. Companion for CallRadar builds everything from the raw `.mp3`s exactly as they come off the phone system — stereo, 8 kHz, agent on the left channel and customer on the right. Every claim it makes is backed by a verified quote, and it measures its own quality the way the 2025 research literature does.

## Results at a glance

Measured on the full 1,441-call run, served live at `/api/evaluation`.

| CALLS ANALYSED | CUSTOMERS · AGENTS | FAITHFULNESS | DIARIZATION |
|---|---|---|---|
| 1,441 | 100 · 10 | 99.9% | 100% |
| the full dataset | | evidence verified | channel separation |

| COVERAGE | RESOLUTION RISK |
|---|---|
| 100% | 195 |
| well-formed outputs | sounded resolved, wasn't |

## The brief, mapped to the implementation

Every requirement in the problem statement is implemented and verified on all 1,441 calls.

| THE BRIEF REQUIRES | ✓ | WHERE |
|---|---|---|
| Speech to text | ✓ | `pipeline/transcribe.py` — faster-whisper |
| Work out who said what | ✓ | Channel-split (agent = L, customer = R) — correct by construction |
| Turn-by-turn with timings | ✓ | Interleaved turns + word-level timings |
| Per customer: name, history, recording + transcript | ✓ | Customers page → `/api/customers` |
| Per call: intent | ✓ | `analysis.intent` |
| Per call: mood + the point where it shifted | ✓ | `analysis.mood.shift.timestamp` + mood timeline |
| Per call: resolution | ✓ | `analysis.resolution.status` |
| Per call: summary (≤ 40 words) | ✓ | Validated ≤ 40 words on 100% of calls |
| Needs a manager's attention, ranked | ✓ | Needs Attention page → `/api/attention` |
| Which issues are trending | ✓ | Trends page → `/api/trends` |
| Per-agent volumes, handle times, outcomes | ✓ | Agents page → `/api/agents` |
| Every judgment cites the moment — timestamp + words | ✓ | Every judgment carries `evidence{timestamp, quote, speaker}` |
| Do not re-transcribe on every request | ✓ | Precomputed once → PostgreSQL; the API only reads |
| Git repo + README to run from scratch | ✓ | This repo; `docker compose run --rm pipeline` |

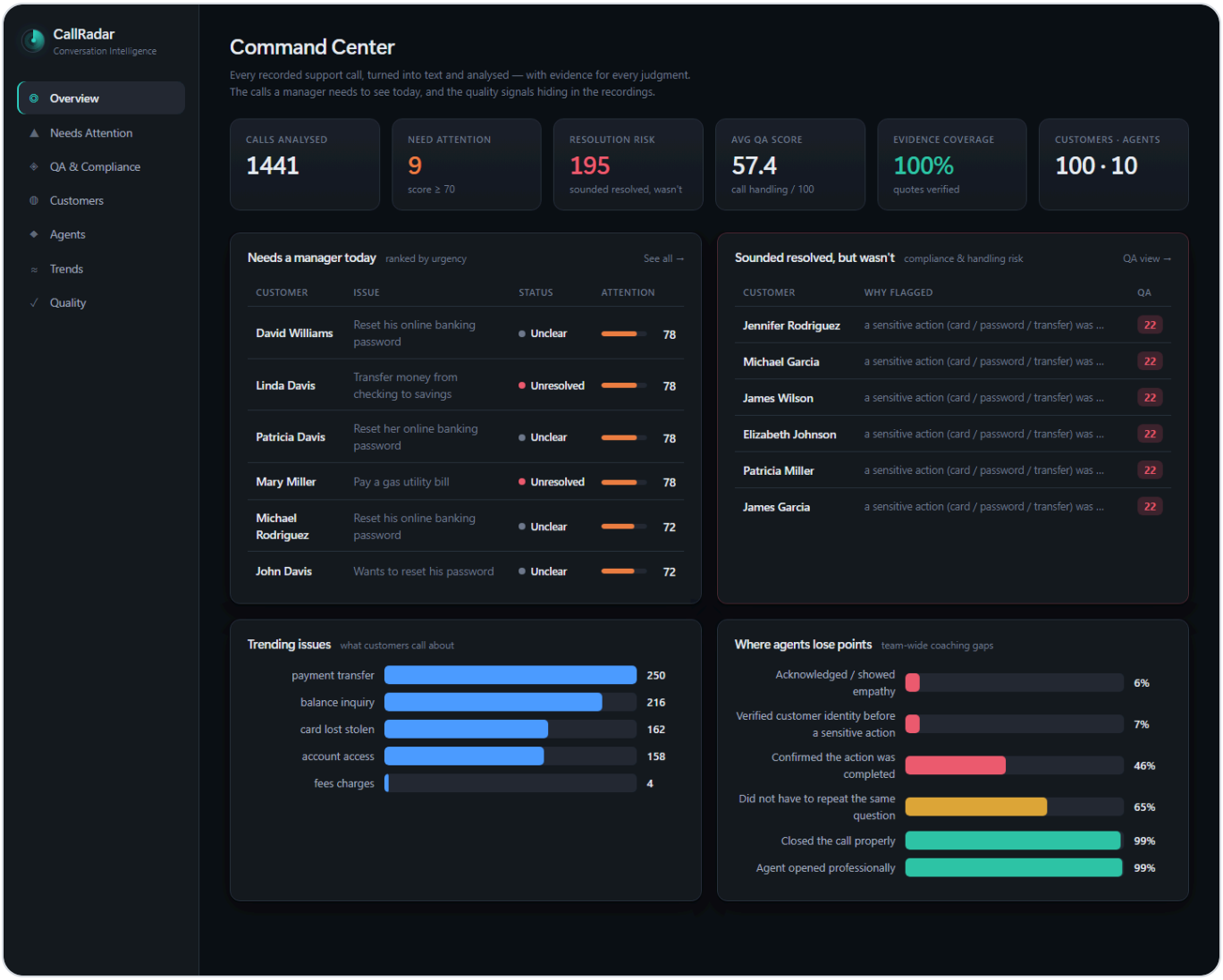
## Why we stand out

Most entries produce a transcript, an LLM summary, and a table. Here's the difference:

| A TYPICAL ENTRY | COMPANION FOR CALLRADAR |
| --- | --- |
| An ML diarizer *guesses* the speakers, and gets some wrong | **Splits the stereo channels** — attribution is correct by construction. Auto-corrects the ~37 reversed-channel recordings |
| The model *says* it's citing a moment; nobody checks | Every quote is **verified against the transcript** at its timestamp — **99.9% faithfulness measured** |
| "It works" / an unverifiable "X% accurate" | We **measure ourselves** the way the 2025 research does — faithfulness, diarization, coverage |
| Summarises what was said | Flags the calls that **sounded resolved but weren't** — money moved with no identity check |
| Needs API keys / a specific cloud | Runs **fully offline**; add Claude/Azure to upgrade |

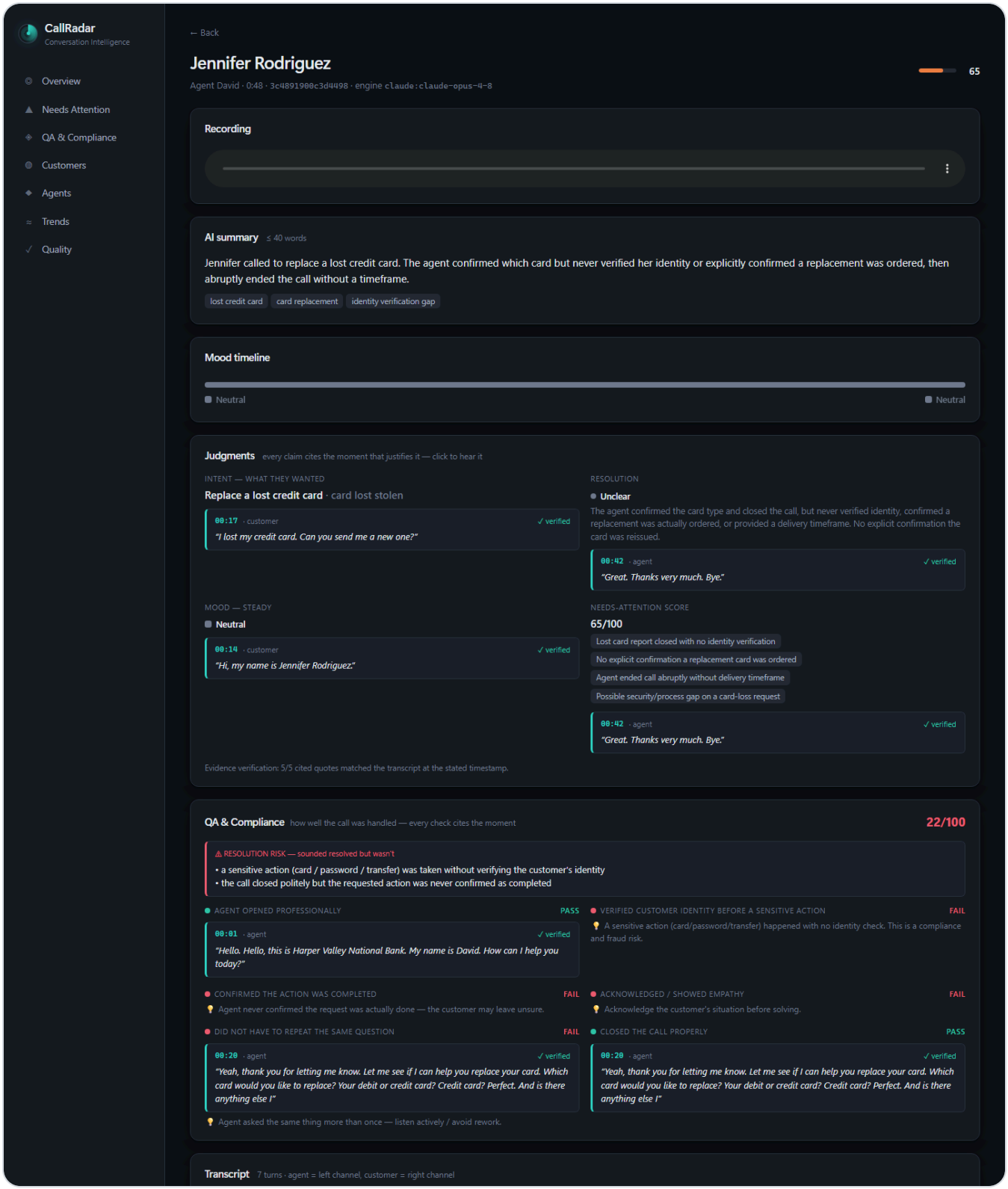
# The dashboard & the API

## Frontend — the manager's Command Center




Headline signals, the calls needing attention today, the "sounded resolved but wasn't" panel, and team-wide coaching gaps.

# Per-call view — recording, transcript, evidence, mood, QA




Playable recording, turn-by-turn transcript, evidence-cited judgments (click a timestamp to hear it), mood timeline, and the QA / compliance breakdown.

## Backend — the required API contract

### GET /api/calls/{sid} — the required per-call contract


```
{
  "sid": "3c4891900c3d4498",
  "customer_name": "Jennifer Rodriguez",
  "agent_name": "David",
  "started_at": "2020-05-30T17:57:47.377000",
  "duration_sec": 47.62,
  "intent_summary": "Replace a lost credit card",
  "intent_category": "card_lost_stolen",
  "mood_start": "neutral",
  "mood_end": "neutral",
  "mood_shifted": false,
  "resolution_status": "unclear",
  "summary": "Jennifer called to replace a lost credit card. The agent confirmed which card but never verified her identity or explicitly confirmed a replacement w
  "attention_score": 65,
  "evidence_score": 1.0,
  "topics": [
    "lost credit card",
    "card replacement",
    "identity verification gap"
  ],
  "analysis_engine": "claude:claude-opus-4-8",
  "qa_score": 22,
  "resolution_risk": true,
  "session_label": "Little Harper Valley 2",
  "ended_at": "2020-05-30T17:58:39.583000",
  "stt_engine": "faster-whisper:base.en",
  "transcript": {
    "turns": [
      {
        "speaker": "agent",
        "start": 1.14,
        "end": 10.77,
        "text": "Hello. Hello, this is Harper Valley National Bank. My name is David. How can I help you today?",
        "words": [
          {
            "start": 1.14,
            "end": 1.64,
            "text": "Hello."
          },
          {
            "start": 6.21,
            "end": 6.61,
            "text": "Hello,"
          },
          {
            "start": 6.93,
            "end": 7.07,
            "text": "this"
          },
          {
            "...": "..."
          }
        ]
      },
      {
        "speaker": "customer",
        "start": 14.74,
        "end": 16.4,
        "text": "Hi, my name is Jennifer Rodriguez.",
        "words": [
          {
            "start": 14.74,
            "end": 15.04,
            "text": "Hi,"
          },
          {
            "start": 15.22,
            "end": 15.34,
            "text": "my"
          },
          {
            "start": 15.34,
            "end": 15.54,
```

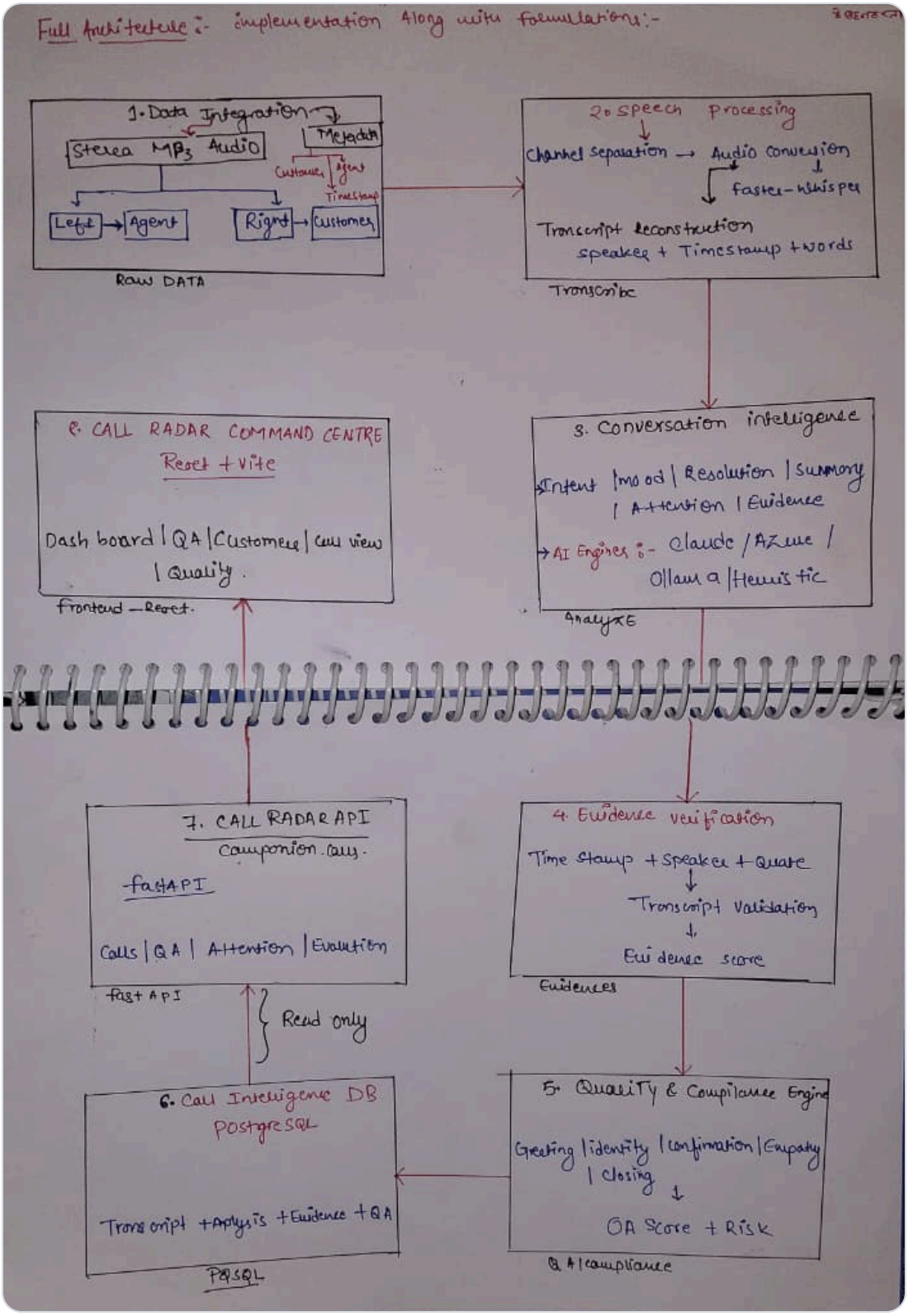

`GET /api/calls/{sid}` returns the transcript with speakers + timings, intent, mood + shift, resolution, ≤40-word summary, 0–100 attention score, QA, and the evidence behind every judgment.

## Architecture

Data flows one way at build time (audio → transcript → analysis → QA → Postgres) and one way at request time (Postgres → API → UI). The two are decoupled: the pipeline can be re-run or swapped to a different engine without touching the serving layer, and the API never pays transcription cost.


## Hand-written design & formulation

The full architecture, the logic behind each stage, and every formula — worked out by hand.




Full architecture: raw audio → transcript → analysis → evidence → QA → DB → API → dashboard.

→ formulation & Notations:-

Diarization - my channels, not guessing

let
'L' → agent
'R' - Customer

we transcribe each separately, giving team sets like 'T_L' and 'T_R'

So transript the final one is like

$$T = \text{Sort}(T\_L \cup T\_R) \text{ order by turn.start}$$

→ separation quality:-

let we prove the channels really are separate speaker by the person Correlation 'ρ' between the two channels' energy envelops "e_L", "e_R" (100ms window):-

Diarization — why channels, not guessing; the transcript reconstruction.

so formulation:-

$$\rho = \frac{\Sigma (e_{Li} - \bar{e}_L)(e_{Ri} - \bar{e}_R)}{\sqrt{[\Sigma (e_{Li} - \bar{e}_L)^2 \cdot \Sigma (e_{Ri} - \bar{e}_R)^2]}}$$

separation_quality = 1 - Max (0, ρ)

Low (even negative) 'ρ' => the two speaker. don't overlap => attribution is correct ** by construction"

measured on this dataset that given

$\rho \approx -0.13$

separation_quality = 1.00 (100%) neatly.

→ Evidence verification -

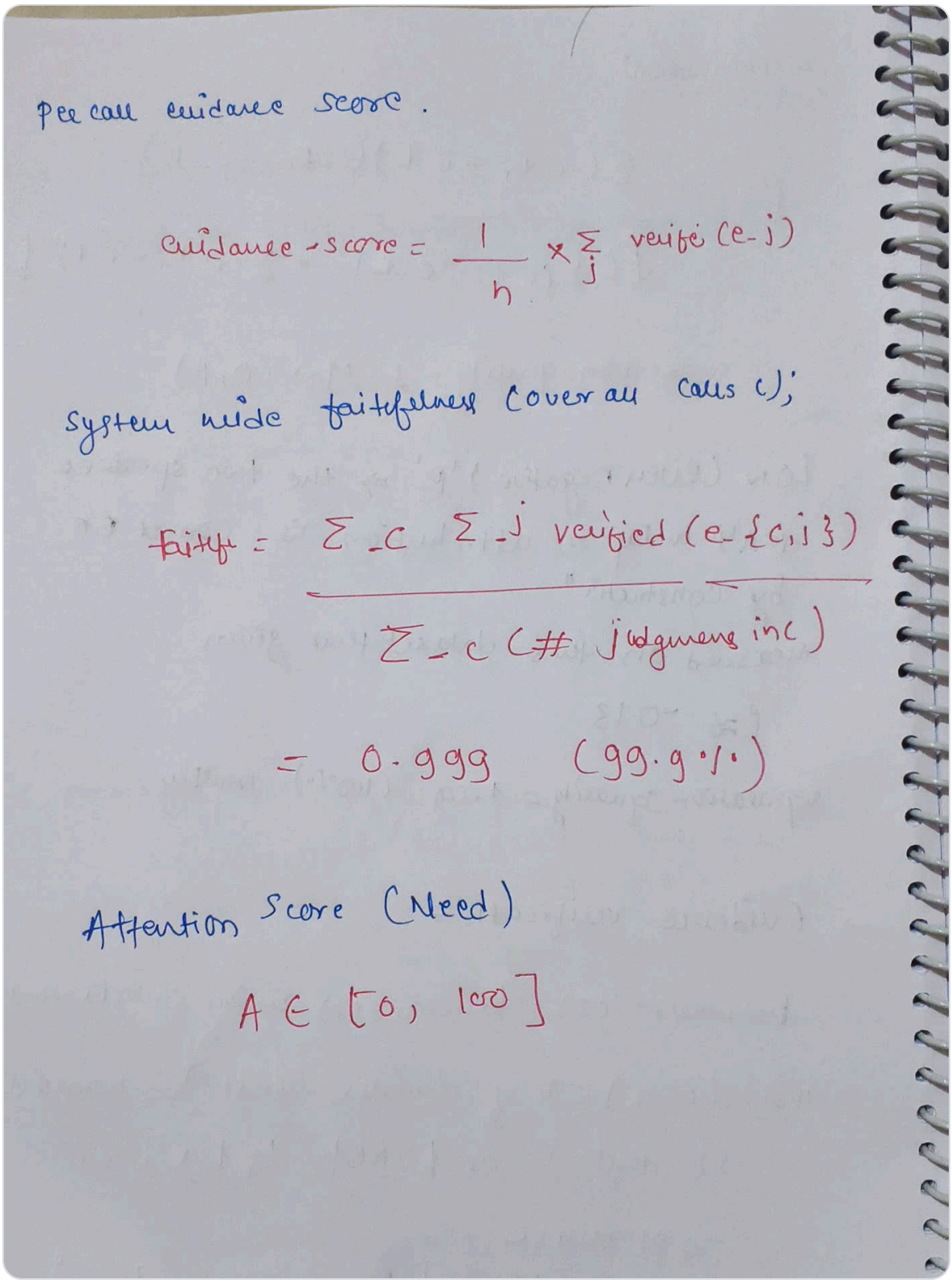
normalise (s) = lowercase(s) with only [a-z0-9]

verified (e_j) = 1 if normalise (quote) ⊆ normalise (z. text) or $|Q \cap B| / |Q| \geq 0.60$

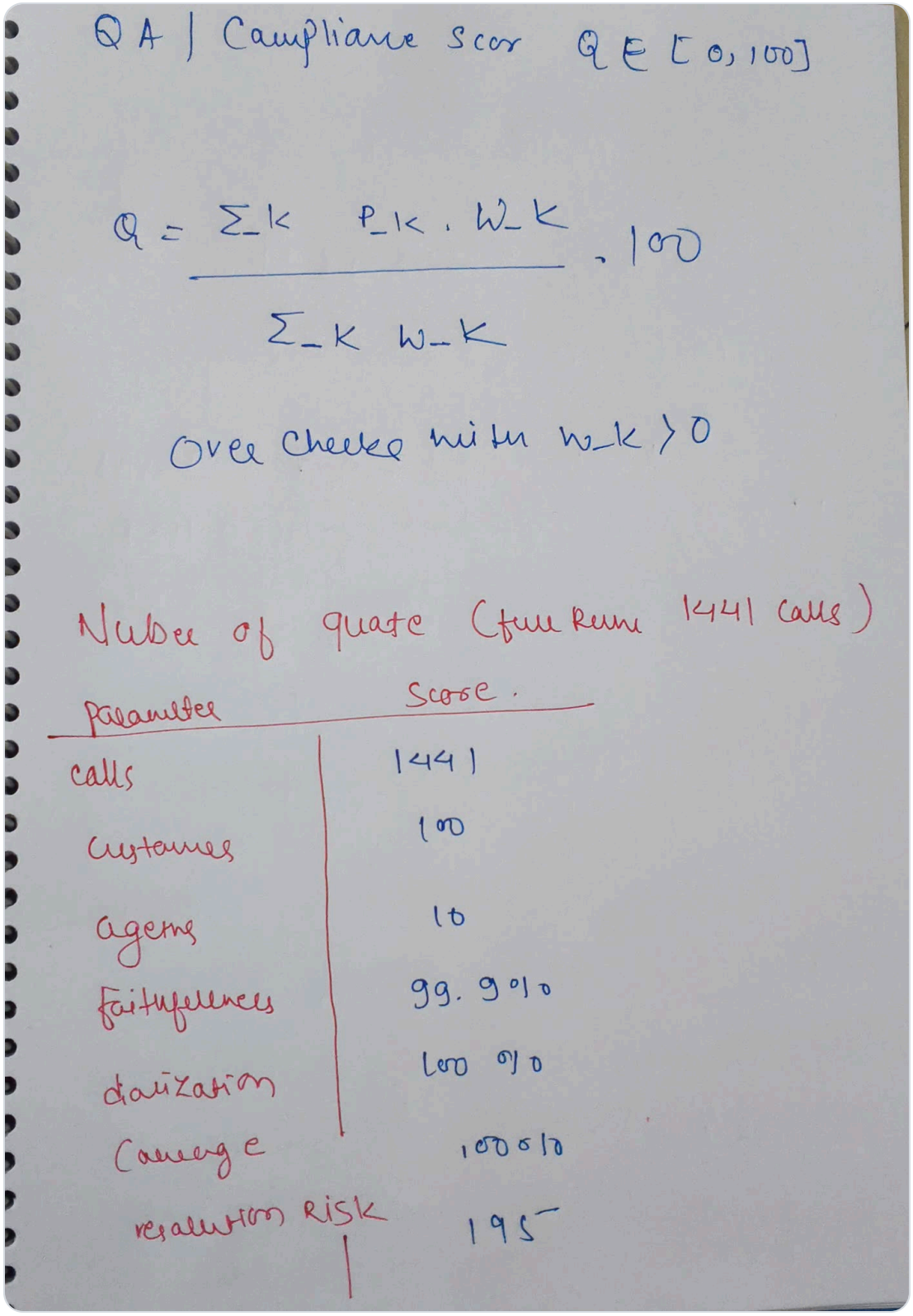
0 otherwise

where Q = words (quote), B = words (z. text)

Pearson correlation ρ, separation quality, and evidence-verification logic.




Per-call evidence score, system-wide faithfulness, and the needs-attention score.




QA / compliance score and the measured results on all 1,441 calls.

## The pipeline — five stages

### 1 · Transcribe (channel-split diarization)

The recordings are stereo — agent left, customer right. Instead of guessing "who spoke" with an ML diarizer, we split the channels and transcribe each independently, then interleave by timestamp. Speaker attribution is correct **by construction**.

```
# c0=c0 = left(agent), c0=c1 = right(customer)
ffmpeg -i call.mp3 -af "pan=mono|c0=c0" -ar 16000 -ac 1 agent.wav
all_turns  = transcribe(agent_wav, "agent")
all_turns += transcribe(cust_wav,  "customer")
all_turns.sort(key=lambda t: t.start)   # rebuild the real conversation
```

### 2 · Analyze (evidence-cited, pluggable engine)

A single tool whose input schema *is* the analysis schema forces the model to return every judgment with an `{timestamp, quote, speaker}` evidence object. Runs with no key (heuristic), or upgrades to Claude / Azure OpenAI.

### 2b · Verify evidence

Each cited quote is checked against the transcript turn at that timestamp (fuzzy, ≥ 60% word overlap). Unverified citations are flagged. This is how we defend against the "wrong evidence scores negative" rule.

### 3 · QA / compliance

A bank-grade rubric scored from the transcript: identity verified before a sensitive action? action confirmed? empathy? repeated question? professional open/close? → a 0–100 QA score and a **resolution-risk** flag for calls that closed politely but were never actually completed.

### 4 · Store & 5 · Serve

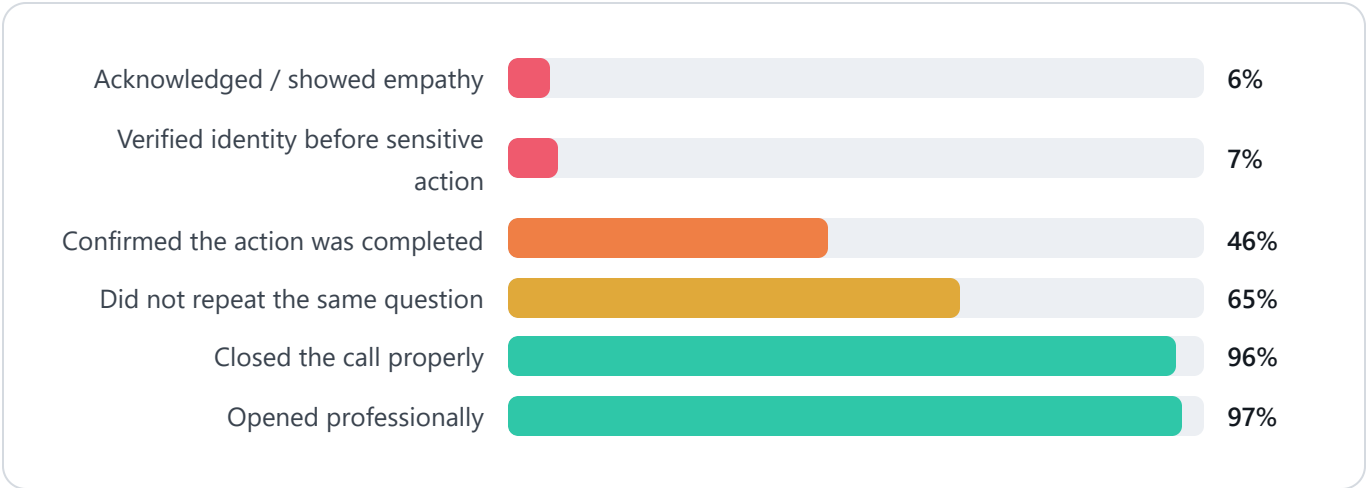
Everything is written once into a PostgreSQL `calls` table. FastAPI only reads — it never re-transcribes on a request.

## Measurement criteria & results

We do **not** claim a single brittle "accuracy %". The conversation-intelligence and LLM-evaluation literature (WER for speech; FaithBench / FaithJudge / LLM-as-a-judge, 2025) measures **faithfulness** — is a claim grounded in the source? — and warns that exact-match scoring punishes good-but-reworded answers. So we measure on the axes that matter:

| CRITERION | WHAT IT MEANS | HOW IT'S COMPUTED | RESULT |
|---|---|---|---|
| **Faithfulness** | Is every judgment backed by a quote that really appears at the cited time? | `verified / total judgments` (≥ 60% word overlap) | 99.9% |
| **Diarization separation** | Are the two speakers genuinely separate? | `1 − max(0, ρ)`, ρ = cross-channel energy correlation | 100% |
| **Coverage** | Is every output well-formed? | summary ≤ 40 words, valid vocab, attention ∈ [0,100], shift ts present, transcript present | 100% |
| **Avg QA score** | How well were calls handled? | weighted compliance checks → 0–100 | 57.4 |
| **Resolution risk** | "Sounded resolved but wasn't" | compliance / confirmation rule | 195 |

### Team-wide coaching gaps (from the QA layer)



## Formulas & notation

### Diarization — separation quality

Let `L` = agent channel, `R` = customer channel; transcribe each into turn sets `T_L`, `T_R`. The final transcript is `T = sort(T_L ∪ T_R)` by `turn.start`. To prove the channels are genuinely separate speakers, take the Pearson correlation `ρ` between the two channels' energy envelopes `e_L, e_R` (100 ms windows):

$$\rho = \frac{\Sigma\,(e\_L_i - \bar{e}\_L)(e\_R_i - \bar{e}\_R)}{\sqrt{\left[\,\Sigma(e\_L_i - \bar{e}\_L)^2 \cdot \Sigma(e\_R_i - \bar{e}\_R)^2\,\right]}}$$

$$\text{separation\_quality} = 1 - \max(0, \rho)$$

$$\text{Measured on this dataset: } \rho \approx -0.13 \rightarrow \text{separation\_quality} = 1.00\ (100\%)$$

Low (even negative) ρ ⇒ the two speakers don't overlap ⇒ attribution is correct by construction.

### Evidence verification & faithfulness

Each judgment `j` carries evidence `e_j = (timestamp, quote, speaker)`. Verification finds the transcript turn `τ` at that timestamp+speaker and accepts if the quote appears in the turn *or* shares ≥ 60% of its significant words:

$$\text{normalise}(s) = \text{lowercase}(s) \text{ keeping only } [a\text{-}z0\text{-}9\ ]$$

$$\text{verified}(e_j) = \begin{cases} 1 & \text{if } \text{normalise}(\text{quote}) \subseteq \text{normalise}(\tau.\text{text}) \\ & \text{or } |Q \cap B| \,/\, |Q| \geq 0.60 \\ 0 & \text{otherwise} \end{cases} \qquad (Q = \text{words}(\text{quote}),\ B = \text{words}(\tau.\text{text}))$$

$$\text{per-call evidence score} = (1/n) \cdot \Sigma_j\ \text{verified}(e_j)$$

$$\text{faithfulness} = \frac{\Sigma_c\ \Sigma_j\ \text{verified}(e_{c,j})}{\Sigma_c\ (\#\ \text{judgments in } c)} = 0.999 \quad (99.9\%)$$

### Needs-attention score A ∈ [0, 100]

```
A = 10                                  (base)
  + min(35, 12·neg)                     (customer negativity)
  + 30 escalated / 28 unresolved / 15 unclear
  + 20 if mood shifted to negative  (−5 if recovered to positive)
  + 22 if HIGH_RISK topic / 10 if MED_RISK
  + 12 if agent repeated a question
  + 10 if dur ≥ 90s  (5 if ≥ 70s)
A = clamp(A, 0, 100)

HIGH_RISK = {fraud_dispute, card_lost_stolen, complaint, account_access}
MED_RISK  = {transaction_issue, fees_charges, loan_mortgage, payment_transfer}
```

### QA / compliance score Q ∈ [0, 100]

Each check `k` has a pass flag `p_k ∈ {0,1}` and a weight `w_k` (identity = 3, action = 2, greeting / empathy / no-repeat / close = 1 each):

```
      Σ_k  p_k · w_k
Q =  ──────────────── × 100        (over checks with w_k > 0)
       Σ_k  w_k

resolution_risk =
   (sensitive_action ∧ ¬identity_verified)                    # compliance
risk
 ∨ (sounded_resolved ∧ customer_asked ∧ ¬action_confirmed ∧ status ≠ resolved)
```

---

## Example output (real, from the API)

A payment/card call that *looks* fine — polite close — but the system catches the compliance failure:

```
{
  "customer_name": "Jennifer Rodriguez",
  "intent_summary": "Replace a lost credit card",
  "mood_start": "neutral", "mood_end": "neutral",
  "resolution_status": "unclear",
  "summary": "Jennifer called to replace a lost credit card. The agent confirmed
              which card but never verified her identity or explicitly confirmed a
              replacement was ordered, then abruptly ended the call.",
  "attention_score": 65,
  "qa_score": 22,
  "resolution_risk": true,
  "evidence_score": 1.0,
  "resolution": {
    "status": "unclear",
    "evidence": { "timestamp": "00:42",
                  "quote": "Great. Thanks very much. Bye.",
                  "speaker": "agent", "verified": true }
  },
  "qa": {
    "resolution_risk_reasons": [
      "a sensitive action (card / password / transfer) was taken without
       verifying the customer's identity"
    ]
  }
}
```


Every field is backed by a verified quote — click any timestamp in the dashboard and it jumps the audio to that exact second.

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## Run it from scratch

Prerequisites: Docker + Docker Compose. Nothing else. Runs with **no API keys**.

```
# 1. Load the dataset into ./data  (accepts the zip OR an extracted folder)
python scripts/load_data.py <callradar-data.zip>

# 2. Bring up Postgres, the API, and the dashboard
docker compose up -d --build

# 3. Turn the recordings into transcripts + analysis  (THE key step)
docker compose run --rm pipeline

# 4. Add the QA / compliance layer + quality metrics
docker compose run --rm pipeline python -m pipeline.compute_qa
docker compose run --rm pipeline python -m pipeline.evaluate

# 5. Open the dashboard → http://localhost:3000   (API → http://localhost:8000)
```

> **Premium analysis (optional):** set `ANTHROPIC_API_KEY` or the `AZURE_OPENAI_*` / `AZURE_CLAUDE_*` variables and run `docker compose run --rm pipeline python -m pipeline.enrich --top 100` to upgrade the highest-attention calls to LLM-grade analysis. The engine is model-agnostic.

## API reference

| ENDPOINT | RETURNS |
| --- | --- |
| `GET /api/calls/{sid}` | Full per-call: transcript (speakers + timings), intent, mood + shift, resolution, ≤40-word summary, attention score, QA, evidence |
| `GET /api/customers` | Every customer with call counts and peak attention |
| `GET /api/customers/{name}` | A customer's full call history |
| `GET /api/attention` | Ranked "needs a manager's attention today" |
| `GET /api/qa` | QA-risk-ranked calls ("sounded resolved but wasn't") |
| `GET /api/qa/stats` | Team-wide QA metrics and coaching gaps |
| `GET /api/evaluation` | System quality: faithfulness, diarization, coverage |
| `GET /api/trends` | Trending issues across all calls |
| `GET /api/agents` | Per-agent volumes, handle times, and outcomes |
| `GET /api/audio/{sid}.mp3` | The playable recording |
| `GET /api/stats` | Dashboard headline numbers |

## Live links & repository

| RESOURCE | LINK |
| --- | --- |
| GitHub repository | github.com/SAHILKOSHRIYA/Companion_Radar_call_01 |
| Dashboard (local) | `http://localhost:3000` |
| API (local) | `http://localhost:8000` |
| Architecture doc | architecture.html · ARCHITECTURE.md |
| Formulas & notation | formulas.html · HANDWRITTEN_REFERENCE.md |
| Demo walkthrough | demo.html · DEMO.md |

**Companion for CallRadar & the Quality Checks** — conversation intelligence over 1,441 consumer-bank support calls.

Team: Sahil Koshriya · Sakshi Stack: faster-whisper · FastAPI · PostgreSQL · React/Vite · Docker Compose. Runs from scratch with no API keys.nbsp;|Stack: faster-whisper · FastAPI · PostgreSQL · React/Vite · Docker Compose. Runs from scratch with no API keys.nbsp; Stack: faster-whisper · FastAPI · PostgreSQL · React/Vite · Docker Compose.